

MEACH — User workflows

Complete guide to every workflow in the MEACH desktop application.

Currency is **Malagasy Ariary (MGA)**, shown with the symbol **Ar** (configurable in Settings).

Table of contents

1. [First launch & login](#)
 2. [Roles & what each role can do](#)
 3. [Shell \(main window\)](#)
 4. [Master data setup \(recommended order\)](#)
 5. [Students](#)
 6. [Attendance](#)
 7. [Billing \(creating fee obligations\)](#)
 8. [Collecting payments & receipts](#)
 9. [Unpaid balances & payment history](#)
 10. [Financial reports](#)
 11. [Administration](#)
 12. [Business rules cheat sheet](#)
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1. First launch & login

What is created automatically

On the **first launch**, MEACH creates a local SQLite database and seeds only what is required to sign in and use the menus:

Seeded	Purpose
Roles	Administrator, Accountant, Cashier, School manager
Admin user	Username admin
Academic levels	L1, L2, L3
Payment types	Droit, Écolage, Livre, Mock Exam, Official Exam
School settings	Default school name, MGA / Ar, receipt footer
Current school year	Derived from today's date (Sept–June)

No students, groups, fees, payments, or demo users are created in a Release build.

If you previously ran a Debug build with demo data, delete the existing database so Release starts clean:

```
%LocalAppData%\SchoolManagement\schooL_management.db
```

Sign in

1. Start MEACH.
2. Enter username and password.
3. Click **Sign in**.

Field	Default administrator
Username	admin
Password	ChangeMe123!

Mandatory password change

The first time `admin` signs in (or after an administrator resets a password with “must change”), a dialog **Choose a new password** appears.

1. Enter the current password.
2. Enter a new password (at least 8 characters, with at least one letter and one digit).
3. Confirm and save.

You cannot use the app until the password is changed. Cancelling signs you out.

Optional password change

From the shell: **Change password** (same rules; not mandatory).

Sign out

Shell → **Logout** → returns to the sign-in window.

2. Roles & what each role can do

Capability	Administrator	Accountant	Cashier	School manager
Dashboard	✓	✓	✓	✓
Students (view)	✓	✓	✓	✓
Students (create / edit / transfer / archive)	✓			✓
Academic levels & groups & schedules	✓			✓
School years (manage)	✓	view		view
Attendance	✓			✓
View fees	✓	✓	✓	✓
Create bills (generate fees)	✓	✓		
Register payments	✓		✓	
Cancel / reverse payments	✓			
Receipts (print)	✓	view	✓	view
Financial reports	✓	✓		✓
Users, payment types, settings, audit	✓			

Menus only appear if your role has permission for that screen.

3. Shell (main window)

Control	Action
Left navigation	Open a module (grouped: Overview, People, Attendance, Payments, Reports, Administration)
Register a payment	Opens the payment dialog (Cashier, Administrator)
Refresh	Reloads the current screen
Change password	Change your own password
Logout	End the session
School year label	Shows the current school year (or a warning if none)

Window title: **MEACH**.

4. Master data setup (recommended order)

Do this once when commissioning a school PC:

Settings → School years → Academic levels → Student groups (+ schedules) → Payment types (if needed) → Students

Settings (Administration → Settings)

Who: Administrator

1. Open **Settings**.
2. Set school name (required on receipts), address, phone, email, website.
3. Confirm currency code **MGA** and symbol **Ar** (or change if needed).
4. Set receipt footer and default due day (1–28).
5. Optionally browse for a logo.
6. **Save**.

Backup database: choose a destination file; MEACH writes a copy of the SQLite database.

Refresh overdue: recomputes overdue status on unpaid fees.

School years (People → School years)

Who: View — Administrator, Accountant, School manager. Manage — Administrator only.

1. **Create** a school year (name, start/end dates).
2. **Set current** on the year that is in use (only open years).
3. When the year ends, **Close** it (blocks new billing and payments for students on that year).
4. **Reopen** if you need to correct data.

Academic levels (People → Academic levels)

Who: School manager, Administrator

1. Review seeded **L1 / L2 / L3**, or create more levels.
2. Select a level to see its groups.
3. **Edit**, **Activate**, or **Deactivate** as needed.

Student groups (People → Student groups)

Who: School manager, Administrator

1. Filter by level / active / search.
2. **Create** a group (name, academic level, optional description).
3. Select the group to see enrolled students and the **schedule**.
4. **Schedules: Add / Edit / Remove** sessions (weekday + start/end time). End must be after start.
5. **Deactivate** or **Archive** a group (archive only if it has no students).

Schedules drive **Daily attendance** (a session must exist for that weekday).

Payment types (Administration → Payment types)

Who: Administrator

Built-in types (seeded): **Droit, Écolage, Livre, Mock Exam, Official Exam**.

1. **Create** custom types (e.g. Excursion, Trip): name, description, default amount, frequency (**OneTime** or **Monthly**).
2. Prefer **OneTime** for types billed under **Other fees**.
3. **Edit** amounts and descriptions; **deactivate** instead of deleting if the type was already used.

Custom types appear under **Payments** → **Other fees** and in the student **Create bill** menu.

5. Students

List (People → Students)

Who: View — most roles. Manage — School manager, Administrator. Create bill — Accountant, Administrator.

1. Search by student number, first name, last name, or **first + last** (either order).
2. Filter by level, group, status; page through results.
3. **Create** — enter identity, contacts, level, group, school year, enrollment date, status. A student number is assigned on save.
4. **Edit** — update the file.
5. **Open** — student detail (fees, payments, receipts, schedule context).
6. **Transfer** — move to another **active** group (level follows the destination group). Optional reason.
7. **Archive** — soft-delete; financial history is kept (confirmation required).
8. **Create bill** (dropdown) — bill this student for Écolage, Droit, Livre, exams, or a custom type.

Student detail

1. Open from the list or from a group's student list.
2. Review identity, fee lines, payments, receipts.
3. **Edit** student (if permitted).
4. **Register a payment** — dialog opens with the student (and optionally a selected fee) prefilled.
5. Open a payment for cancel/reverse (Administrator) or receipt details.
6. **Back** returns to the student list.

Transfer workflow

1. Select student → **Transfer**.
 2. Choose destination group (must be active, different from current).
 3. Confirm. Academic level updates to match the group.
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6. Attendance

Daily attendance (Attendance → Daily attendance)

Who: School manager, Administrator

Prerequisites: Group with a **class schedule** for the weekday of the chosen date.

1. Select academic level → student group → date.
2. **Load sheet** — students of that group appear.
3. Set status per student: Present / Absent / Late / Excused; optional remarks.
4. Shortcuts: mark all present, or mark selected absent.
5. **Save**.

If no session is scheduled that weekday, save fails with a clear error.

Attendance reports (Attendance → Attendance reports)

1. Filter by level, group, date range.
 2. **Run** to see per-student presence statistics.
 3. Read-only (no export on this screen).
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7. Billing (creating fee obligations)

Billing creates **StudentFee** lines (what the student owes). Collecting money is a separate step (section 8).

Who can bill: Accountant, Administrator ([ManageFees](#)).

Shared options on one-time and monthly bill dialogs:

- **Entire group / all groups** or **One student**
- School year (must be **open**)
- Amount and due date / due day
- Optional "only active enrolments" when billing a group

Droit (Payments → Droit)

1. Click **Bill Droit**.
2. Choose school year, amount (defaults from payment type), due date, target students.
3. Confirm.

Rule: At most **one Droit per student per school year**. Existing lines are skipped.

Écolage (Payments → Écolage)

1. Click **Generate monthly fees**.
2. Choose school year, months within the year, amount per month, due day (1–28).
3. Target: one student or a group / all groups.
4. Confirm.

Rule: One Écolage line per student per **month + year**. Already billed months are skipped.

Livre (Payments → Livre)

1. Click **Bill Livre**.
2. Enter **Book description** (required), school year, amount, due date, target.
3. Confirm.

Rule: Many Livre lines per year are allowed. The same description in the same year for the same student is skipped.

Mock Exam / Official Exam

Same pattern as Livre: description required; many lines per year, distinguished by description.

Other fees (Payments → Other fees)

For **custom** payment types only (not the five built-ins).

1. Click **Create bill**.
2. Select the payment type (e.g. Excursion, Trip) — amount fills from the type's default and updates when you change type.
3. Enter description, school year, due date, target.
4. Confirm.

Rule: Same description uniqueness as Livre (many lines per year when descriptions differ).

Create bill from the student list

Same dialogs as above, locked to the selected student. Useful for one-off billing without opening the fee screen.

8. Collecting payments & receipts

Register a payment

Who: Cashier, Administrator

Entry points: Shell button · fee lists · unpaid balances · payment history · student detail

Prerequisites: At least one outstanding fee; student's school year is **open**.

1. Select the student (or keep the prefilled one).
2. Outstanding obligations load — pick the fee line to settle.
3. Amount defaults to the **remaining** balance (partial payments allowed; cannot exceed remaining).
4. Set payment date, method (Cash, Bank transfer, Mobile money, Cheque, Other), optional reference and notes.
5. Leave **print receipt** on if you want a print after save.
6. Confirm.

Payment and receipt are saved together. A possible duplicate (same student / date / amount) shows a confirmation before continuing. If printing fails, the payment still remains registered.

Receipts (Payments → Receipts)

Who: View — roles with receipt access. Print — Cashier, Administrator

1. Filter by date range and search (receipt number, payment number, student number/name).
2. **Print** or **Export PDF**.
3. Open the linked payment if needed.

Each print increases the print count (reprints remain identifiable).

9. Unpaid balances & payment history

Unpaid balances (Payments → Unpaid balances)

1. Opens filtered to **outstanding** fees.
2. Filter, search (including full name), export PDF/Excel.
3. **Register payment** on a selected line.
4. **Refresh overdue** recalculates overdue flags for past-due unpaid lines.

Payment history (Payments → Payment history)

1. Filter by dates, level, group, type, method, status, cashier (when available).
2. Shortcut for **today**.
3. Register payment, open payment detail, export.

Statuses: **Active**, **Cancelled**, **Reversed**. Page totals count **Active** payments only.

Cancel or reverse a payment

Who: Administrator only

1. Open payment detail (from history, receipts, or student file).
2. Choose **Cancel** or **Reverse**.
3. Confirm and enter a reason.

Only **Active** payments can be cancelled/reversed. Fee balances are restored; the payment remains in history with the new status. The school year must be open.

10. Financial reports

Who: Accountant, School manager, Administrator

Screen: Reports → Financial reports

Section	Content
Daily payments	Cash of a single day, by method
Monthly financial	Expected / collected / remaining over a period
By class	Breakdown by student group
By payment type	Breakdown by fee type
Outstanding balances	What is still owed

1. Set filters (level, group, type; single date for Daily; range otherwise).
 2. Results refresh when filters change.
 3. **Export PDF** or **Excel**.
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11. Administration

Users

Who: Administrator

1. Search by username, name (including first + last), or email.
2. **Create** user: name, username, password, role, optional “must change password”.
3. **Edit** profile and role.
4. **Reset password** (can force change at next login).
5. **Deactivate** / **Reactivate** (accounts are not deleted — they may appear on payment history). You cannot deactivate yourself.

Audit log

Who: Administrator

Filter by action, user, date range, and search text. Read-only trail of logins, billing, payments, cancellations, and configuration changes.

Dashboard (Overview)

Read-only snapshot: student counts (incl. by level), groups, collections, outstanding/overdue, absences today, recent payments, collected-by-type. Optionally scoped by school year where the UI offers it.

12. Business rules cheat sheet

Fee type	How many per student	Description
Droit	One per school year	Not used to tell lines apart
Écolage	One per month + year	N/A
Livre	Many per year	Required — unique per year
Mock / Official Exam	Many per year	Required — unique per year
Custom (Other fees)	Many per year	Required — unique per year

- Closed school year → no new bills; no register / cancel / reverse payments for students on that year.
 - Financial records are cancelled or reversed, not hard-deleted.
 - Fee list screens share filters, summary totals, PDF/Excel export, **Register payment**, and **Open student**.
 - Name search accepts a single name or **First Last** / **Last First** (case-insensitive) on students, fees, payments, receipts, and users.
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Quick start checklist (empty school)

1. Sign in as `admin` → change password.
2. **Settings** — school name and identity.
3. Confirm **School year** (current).
4. Confirm **L1 / L2 / L3** (or create levels).
5. Create **student groups** and **schedules**.
6. Adjust **payment type** default amounts.
7. Enrol **students**.
8. **Bill** Droit / Écolage / other fees as needed.

9. **Register payments** and print **receipts**.
10. Use **Reports** and schedule regular **database backups**.